Chap 2. The Space of Rough Paths

Moon, Bora

March 20th, 2020

Seoul National University

Outline

1. Review
2. Main contents
   - Basic definitions

     -The space of α-Hölder rough paths

     -The space of geometric rough paths
   - Rough paths as Lie group valued paths
3. Conclusion

Review

Review: Definitions

Let $\alpha > 0$ and E be a Banach space.

Definition. A function $X : [0, T] \to E$ is α-Hölder continuous if

$$\|X\|_\alpha := \sup_{s \neq t \in [0,T]} \frac{|X_{s,t}|_E}{|t-s|^\alpha} < \infty.$$

The space of all α-Hölder continuous functions from $[0, T]$ into E is denoted by $\mathcal{C}^\alpha([0, T]; E) = \mathcal{C}^\alpha(E) = \mathcal{C}^\alpha$.

Notation.

1. $X_{s,t} := X_t - X_s$ is the path increment.

2. For a function $\Xi : [0, T]^2 \to E$, we write $\Xi \in \mathcal{C}_2^\alpha([0, T]^2; E)$ if

$$\|\Xi\|_\alpha := \sup_{s \neq t \in [0,T]} \frac{|\Xi_{s,t}|_E}{|t-s|^\alpha} < \infty, \quad \text{where} \quad \Xi_{s,t} := \Xi(s, t).$$

Remark.

1. If $X \in \mathcal{C}^\alpha([0,T];E)$, then its increment $X_{s,t} \in \mathcal{C}_2^\alpha([0,T]^2;E)$.

2. The quantity $\|\cdot\|_\alpha$ is a semi-norm; however, the quantity

$$\|X\|_{\mathcal{C}_\alpha} := |X_0|_E + \|X\|_\alpha,$$

is a norm that makes $\mathcal{C}^\alpha(E)$ a Banach sapce (in general not separable).

3. If $\alpha < \beta$, then $\mathcal{C}^\beta \subset \mathcal{C}^\alpha$.

Remark.

1. If $X \in \mathcal{C}^\alpha([0,T];E)$, then its increment $X_{s,t} \in \mathcal{C}_2^\alpha([0,T]^2;E)$.

2. The quantity $\|\cdot\|_\alpha$ is a semi-norm; however, the quantity

$$\|X\|_{\mathcal{C}_\alpha} := |X_0|_E + \|X\|_\alpha,$$

is a norm that makes $\mathcal{C}^\alpha(E)$ a Banach sapce (in general not separable).

* When working with paths starting at the origin, the term $|X_0|_E$ can be omitted: we can work with directly with $\|\cdot\|_\alpha$.

* The same is true if we are interested in the α-Hölder metric between two paths started at the same point.

3. If $\alpha < \beta$, then $\mathcal{C}^\beta \subset \mathcal{C}^\alpha$.

Remark.

1. If $X \in \mathcal{C}^\alpha([0,T];E)$, then its increment $X_{s,t} \in \mathcal{C}_2^\alpha([0,T]^2;E)$.

2. The quantity $\|\cdot\|_\alpha$ is a semi-norm; however, the quantity

$$\|X\|_{\mathcal{C}_\alpha} := |X_0|_E + \|X\|_\alpha,$$

   is a norm that makes $\mathcal{C}^\alpha(E)$ a Banach sapce (in general not separable).

3. If $\alpha < \beta$, then $\mathcal{C}^\beta \subset \mathcal{C}^\alpha$.

Review: Motivation

- Main object: differential equation of the form

$$dY_t = f(Y_t)dX_t. \tag{1}$$

  - E and V : Banach spaces
  - $X : [0, T] \to E$, input; $\quad Y : [0, T] \to V$, output (unknown)
  - $f : V \to \mathcal{L}(E; V)$ is a smooth function
  - $\mathcal{L}(E; V) :=$ the space of continuous linear maps from E to V.

- Main object: differential equation of the form

$$dY_t = f(Y_t)dX_t. \tag{1}$$

  - E and V : Banach spaces
  - $X : [0, T] \to E$, input; $\quad Y : [0, T] \to V$, output (unknown)
  - $f : V \to \mathcal{L}(E; V)$ is a smooth function
  - $\mathcal{L}(E; V) :=$ the space of continuous linear maps from E to V.

* Common choice : $E = \mathbb{R}^d$ and $V = \mathbb{R}^n$, so that $f : \mathbb{R}^n \to \mathbb{R}^{n \times d}$.

* We expect X and Y to be continuous functions.

Review: Motivation

- Main object: differential equation of the form

$$dY_t = f(Y_t)dX_t. \tag{1}$$

  - E and V : Banach spaces
  - $X : [0, T] \to E$, input; $Y : [0, T] \to V$, output (unknown)
  - $f : V \to \mathcal{L}(E; V)$ is a smooth function
  - $\mathcal{L}(E; V) :=$ the space of continuous linear maps from E to V.

- Usual way to interpret equation (1) with initial data y_0: integral form

$$Y_t = y_0 + \int_0^t f(Y_s)dX_s. \tag{2}$$

- Usual way to interpret equation (1) with initial data y_0: integral form

$$Y_t = y_0 + \int_0^t f(Y_s) dX_s. \tag{2}$$

  - $X, Y \in \mathcal{C}^\alpha$ (and hence $t \mapsto f(Y_t)$ is α-Hölder as well)

Remark.

1. If $\alpha + \alpha > 1$, we can interpret the integral appearing in (2) as a Young integral and find the solution in the space $\mathcal{C}^\alpha$.

2. If $\alpha + \alpha \leq 1$, the question is more tricky; the theory of rough paths provides a convenient answer and many classical results about stochastic differential equations are recovered in rough paths theory.

Observation

- Suppose that $X \in \mathcal{C}^\alpha([0,T];\mathbb{R}^d)$, $\alpha \in (\frac{1}{3}, \frac{1}{2}]$,

and that $Y : [0,T] \to \mathbb{R}^n$ solves (1) with smooth $f : \mathbb{R}^n \to \mathbb{R}^{n\times d}$:

$$dY_t = f(Y_t) dX_t.$$

- We expect that, at least on small scales,

$$Y_{s,t} = f(Y_s) X_{s,t} + R(s,t),$$

where R is some remainder; and thus

$$f(Y_t) - f(Y_s) = Df(Y_s) Y_{s,t} + \tilde{R}(s,t) = Df(Y_s) f(Y_s) X_{s,t} + \hat{R}(s,t)$$

for suitable remainder terms $\tilde{R}$ and $\hat{R}$.

Observation

- Neglecting the remainder and setting $Z_s := Df(Y_s)f(Y_s)$, we have

$$\begin{aligned}\int_0^t f(Y_s)dX_s &= \int_0^t f(Y_0) + (f(Y_s) - f(Y_0))dX_s \\ &\approx f(Y_0)X_{0,t} + Z_0 \int_0^t X_{0,s} \otimes dX_s.\end{aligned}$$

Note. Since $\alpha \leq \frac{1}{2}$, this is again an ill-posed task: Rough path theory proposes to *postulate* the value of

$$\int_s^t X_{s,r} \otimes dX_r := \mathbb{X}_{s,t}, \qquad (*)$$

and to regard a path as not only X but a couple $(X, \mathbb{X})$.

Basic Definitions

The space of α-Hölder rough paths

Definition. For $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ and a Banach space V, the space $\mathscr{C}^\alpha([0,T],V)$ of α-Hölder rough paths is the space of pairs

$$\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha([0,T];V) \oplus \mathcal{C}_2^{2\alpha}([0,T]^2; V \otimes V),$$

satisfying the algebraic relation (Chen's relation):

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = X_{s,u} \otimes X_{u,t}, \quad \forall s,u,t \in [0,T].$$

- Chen's relation:

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = X_{s,u} \otimes X_{u,t}, \quad \forall s, u, t \in [0, T].$$

Remark.

1. Since $X_{t,t} = X_t - X_t = 0$, $\mathbb{X}_{t,t} = 0$, for all t.

2. This algebraic condition is natural:

If X is a smooth function and we read (*) from right to left, that is,

$$\mathbb{X}_{s,t} := \int_s^t X_{s,r} \otimes dX_r \left(= \int_s^t X_{s,r} \otimes \dot{X}_r d_r \right),$$

then it follows that

$$\begin{aligned}
&\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} \\
&= \int_s^t X_{s,r} \otimes dX_r - \int_s^u X_{s,r} \otimes dX_r - \int_u^t X_{u,r} \otimes dX_r \\
&= X_{s,u} \otimes \int_u^t dX_r = X_{s,u} \otimes X_{u,t}.
\end{aligned}$$

Analytic condition

Q. If X is assumed to be α-Hölder continuous, then what is the natural analytic condition that should be imposed on $\mathbb{X}$?

A. Since X is α-Hölder continuous,

$$|X_{s,t}| \lesssim |t-s|^{\alpha} \quad \Rightarrow \quad X_{\lambda s, \lambda t} \sim \lambda^{\alpha} X_{s,t}.$$

Considering (*), it is natural that $\mathbb{X}$ is also self-similar with

$$\mathbb{X}_{\lambda s, \lambda t} \sim \lambda^{2\alpha} \mathbb{X}_{s,t}.$$

- Analytic condition:

$$\|X\|_{\alpha} := \sup_{x \neq s \in [0,T]} \frac{|X_{s,t}|}{|t-s|^{\alpha}} < \infty, \ \|\mathbb{X}\|_{2\alpha} := \sup_{x \neq s \in [0,T]} \frac{|\mathbb{X}_{s,t}|}{|t-s|^{2\alpha}} < \infty.$$

Rough path norm and metric

• If we ignore Chen's relation, the space $\mathscr{C}^{\alpha}$ is naturally embedded in the Banach space $\mathcal{C}^{\alpha} \oplus \mathcal{C}_2^{2\alpha}$ with (semi)-norm $\|X\|_{\alpha} + \|\mathbb{X}\|_{2\alpha}$.

However, due to the nonlinear constraint given by Chen's relation, the space $\mathscr{C}^{\alpha}$ is a closed set in $\mathcal{C}^{\alpha} \oplus \mathcal{C}_2^{2\alpha}$ but is not a linear space.

▶ On $\mathscr{C}^{\alpha}([0, T], V)$, we define the rough path norm

$$|||\mathbf{X}|||_{\alpha} := \|X\|_{\alpha} + \sqrt{\|\mathbb{X}\|_{2\alpha}},$$

which is not a norm in the common sence (the space is not linear), but is homogeneous with respect to the natural scaling $(X, \mathbb{X}) \mapsto (\lambda X, \lambda^2 \mathbb{X})$.

Rough path norm and metric

- Given rough paths $\mathbf{X}, \mathbf{Y} \in \mathscr{C}^\alpha([0, T], V)$, we define the (inhomogeneous) α-Hölder rough path metric

$$\varrho_\alpha(\mathbf{X}, \mathbf{Y}) := \sup_{s \neq t \in [0,T]} \frac{|X_{s,t} - Y_{s,t}|_V}{|t-s|^\alpha} + \sup_{s \neq t \in [0,T]} \frac{|\mathbb{X}_{s,t} - \mathbb{Y}_{s,t}|_{V \otimes V}}{|t-s|^{2\alpha}}.$$

Remark.

1. $(\mathscr{C}^\alpha, \varrho_\alpha)$ is a complete metric space.
2. If $\frac{1}{3} < \alpha < \beta \leq \frac{1}{2}$, then $\mathscr{C}^\beta \subset \mathscr{C}^\alpha$.

Remark

1. Given an arbitrary path $X \in \mathcal{C}^\alpha(V)$, it doesn't seem obvious that this path can indeed be lifted to a rough path $(X, \mathbb{X})$.

Lyons-Victoir Extension Theorem:

- If $\frac{1}{3} < \alpha < \frac{1}{2}$, given $X \in \mathcal{C}^\alpha(V)$, there is an associated rough path $(X, \mathbb{X})$. But the choice of $\mathbb{X}$ is not unique.
- For $\alpha = \frac{1}{2}$, it holds only when V is finite-dimensional.

Remark

2. Non-uniqueness of $\mathbb{X}$:

Chen's relation is not enough to determine the unigue $\mathbb{X}$ for X:

$\because$ If we set $\tilde{\mathbb{X}}_{s,t} := \mathbb{X}_{s,t} + F_t - F_s$ for any continuous F, then pair $(X, \tilde{\mathbb{X}})$ satisfies Chen's relation.

$\mathbb{X}$ is determined up to increments of a 2α-Hölder continuous function F.

In general, there is no "canonical" choice for F.

Remark

3. Knowledge of the path $t \mapsto (X_{0,t}, \mathbb{X}_{0,t})$ determines the entire second order process $\mathbb{X}_{s,t}$.

If $(X, \mathbb{X})$ and $(X, \tilde{\mathbb{X}})$ are both rough paths, then by Chen's relation we have

$$\mathbb{X}_{s,t} - \tilde{\mathbb{X}}_{s,t} = \mathbb{X}_{0,t} - \tilde{\mathbb{X}}_{0,t} - \mathbb{X}_{0,s} + \tilde{\mathbb{X}}_{0,s}.$$

The space of geometric rough paths

• Suppose that X is smooth function taking values in $\mathbb{R}^d$ and set $\mathbb{X}_{s,t} := \int_s^t X_{s,r} \otimes dX_r$. Then, we have

$$\begin{aligned}
\mathbb{X}^{i,j}_{s,t} + \mathbb{X}^{j,i}_{s,t} &= \int_s^t X^i_{s,r} dX^j_r + \int_x^t X^j_{s,r} dX^i_r \\
&= \int_s^t d(X^i X^j)_r - X^i_s X^j_{s,t} - X^j_s X^i_{s,t} \\
&= (X^i X^j)_{s,t} - X^i_s X^j_{s,t} - X^j_s X^i_{s,t} = X^i_{s,t} X^j_{s,t}
\end{aligned}$$

and the symmetric part of $\mathbb{X}$ is given by

$$\mathrm{Sym}(\mathbb{X}_{s,t}) = \frac{1}{2} X_{s,t} \otimes X_{s,t}. \tag{4}$$

Definition. The set of geometric rough paths $\mathscr{C}^\alpha_g$ consists of those $(X, \mathbb{X}) \in \mathscr{C}^\alpha$ that satisfy equation (4).

Definition. The set of geometric rough paths $\mathscr{C}_g^{\alpha}$ consists of those $(X, \mathbb{X}) \in \mathscr{C}^{\alpha}$ that satisfy equation (4).

Remark

1. Equation (4) holds for smooth paths considered as rough paths with their canonical lift $\mathbb{X}_{s,t} = \int_s^t X_{s,r} \otimes dX_r$.

$\mathscr{C}_g^{0,\alpha}$:the closure of the set of these canonical lifts of smooth paths in $\mathscr{C}^{\alpha}$

2
- $\mathscr{C}_g^{0,\alpha}$ is strictly smaller than $\mathscr{C}_g^{\alpha}$.
- If $\frac{1}{3} < \alpha < \beta \leq \frac{1}{2}$, then $\mathscr{C}_g^{\beta} \subset \mathscr{C}_g^{0,\alpha}$.

Rough paths as Lie-group valued paths

Rough paths as Lie-group valued paths

- It is natural to see an α-rough path $(X, \mathbb{X}) \in \mathscr{C}^\alpha([0, T], \mathbb{R}^d)$, $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ as a function

$$[0, T]^2 \longmapsto \mathbb{T}^{(2)} = \mathbb{T}^{(2)}(\mathbb{R}^d) := \mathbb{R} \oplus \mathbb{R}^d \oplus (\mathbb{R}^d \otimes \mathbb{R}^d)$$
$$(s, t) \longrightarrow (1, X_{s,t}, \mathbb{X}_{s,t}) =: \mathscr{X}_{s,t}.$$

- Define a product $\otimes$ on $\mathbb{T}^{(2)}$ as

$$(a, b, c) \otimes (a', b', c') = (aa', ab' + a'b, ac' + a'c + b \otimes b').$$

Rough paths as Lie-group valued paths

- It is natural to see an α-rough path $(X, \mathbb{X}) \in \mathscr{C}^\alpha([0,T], \mathbb{R}^d)$, $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ as a function

$$\begin{aligned} [0,T]^2 &\longmapsto \mathbb{T}^{(2)} = \mathbb{T}^{(2)}(\mathbb{R}^d) := \mathbb{R} \oplus \mathbb{R}^d \oplus (\mathbb{R}^d \otimes \mathbb{R}^d) \\ (s,t) &\longrightarrow (1, X_{s,t}, \mathbb{X}_{s,t}) =: \mathscr{X}_{s,t}. \end{aligned}$$

- Define a product $\otimes$ on $\mathbb{T}^{(2)}$ as

$$(a,b,c) \otimes (a',b',c') = (aa', ab' + a'b, ac' + a'c + b \otimes b').$$

Check. $(1,0,0)$: unit element and

$$(1, X_{s,t}, \mathbb{X}_{s,t}) = (1, X_{s,u}, \mathbb{X}_{s,u}) \otimes (1, X_{u,t}, \mathbb{X}_{u,t}), \quad \forall s,u,t \in [0,T].$$

Rough paths as Lie-group valued paths

- It is natural to see an α-rough path $(X, \mathbb{X}) \in \mathscr{C}^\alpha([0, T], \mathbb{R}^d)$, $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ as a function

$$\begin{aligned}[0, T]^2 &\longmapsto \mathbb{T}^{(2)} = \mathbb{T}^{(2)}(\mathbb{R}^d) := \mathbb{R} \oplus \mathbb{R}^d \oplus (\mathbb{R}^d \otimes \mathbb{R}^d) \\ (s, t) &\longrightarrow (1, X_{s,t}, \mathbb{X}_{s,t}) =: \mathscr{X}_{s,t}.\end{aligned}$$

- Define a product $\otimes$ on $\mathbb{T}^{(2)}$ as

$$(a, b, c) \otimes (a', b', c') = (aa', ab' + a'b, ac' + a'c + b \otimes b').$$

Check. $(1, 0, 0)$: unit element and

$$(1, X_{s,t}, \mathbb{X}_{s,t}) = (1, X_{s,u}, \mathbb{X}_{s,u}) \otimes (1, X_{u,t}, \mathbb{X}_{u,t}), \quad \forall s, u, t \in [0, T].$$

($\because$ $X_{s,t} = X_{s,u} + X_{u,t}$ and Chen's relation)

Rough paths as Lie-group valued paths

- It is natural to see an α-rough path $(X, \mathbb{X}) \in \mathscr{C}^\alpha([0, T], \mathbb{R}^d)$, $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ as a function

$$\begin{aligned} [0, T]^2 &\longmapsto \mathbb{T}^{(2)} = \mathbb{T}^{(2)}(\mathbb{R}^d) := \mathbb{R} \oplus \mathbb{R}^d \oplus (\mathbb{R}^d \otimes \mathbb{R}^d) \\ (s, t) &\longrightarrow (1, X_{s,t}, \mathbb{X}_{s,t}) =: \mathscr{X}_{s,t}. \end{aligned}$$

- Define a product $\otimes$ on $\mathbb{T}^{(2)}$ as

$$(a, b, c) \otimes (a', b', c') = (aa', ab' + a'b, ac' + a'c + b \otimes b').$$

- Chen's relation:

$$\mathscr{X}_{s,t} = \mathscr{X}_{s,u} \otimes \mathscr{X}_{u,t}.$$

Rough paths as Lie-group valued paths

- Set

$$\mathbb{T}_a^{(2)}(\mathbb{R}^d) := \{a\} \oplus \mathbb{R}^d \oplus (\mathbb{R}^d \otimes \mathbb{R}^d)$$

Check. For each $(1, b, c) \in \mathbb{T}_1^{(2)}(\mathbb{R}^d)$,

$$(1, b, c) \otimes (1, -b, -c + b \otimes b) = (1, -b, -c + b \otimes b) \otimes (1, b, c) = (1, 0, 0)$$

- $\mathbb{T}_1^{(2)}$ is a group with respect to the operation $\otimes$.
- A rough path $(X, \mathbb{X})$ can be though of as representing the increments of the group-valued path $t \mapsto (1, X_{0,t}, \mathbb{X}_{0,t}) =: \mathscr{X}_{0,t}$:

$$\mathscr{X}_{s,t} = \mathscr{X}_{0,s}^{-1} \otimes \mathscr{X}_{0,t}.$$

Rough paths as Lie-group valued paths

- Using the identifying

$$1 \leftrightarrow (1,0,0), \quad b \leftrightarrow (0,b,0), \quad c \leftrightarrow (0,0,c),$$

and the "power series"

$$\log(1+x) = x - \frac{x^2}{2} + \cdots, \quad \exp(x) = 1 + x + \frac{x^2}{2!} + \cdots,$$

we define the logarithmic map and exponetial map as

$$\log(1+b+c) := b + c - \frac{1}{2}(b+c)\otimes(b+c) = b + c - \frac{1}{2} b \otimes b,$$
$$\exp(b+c) := 1 + (b+c) + \frac{1}{2}(b+c)\otimes(b+c) = 1 + b + c + \frac{1}{2} b \otimes b.$$

Rough paths as Lie-group valued paths

- Using the identifying

$$1 \leftrightarrow (1,0,0), \quad b \leftrightarrow (0,b,0), \quad c \leftrightarrow (0,0,c),$$

we define the logarithmic map and exponetial map as

$$\log(1+b+c) := b + c - \frac{1}{2} b \otimes b,$$
$$\exp(b+c) := 1 + b + c + \frac{1}{2} b \otimes b.$$

-

$$\mathbb{T}_0^{(2)}(\mathbb{R}^d) = \{0\} \oplus \mathbb{R}^d \oplus (\mathbb{R}^d \otimes \mathbb{R}^d) \cong \mathbb{R}^d \oplus \mathbb{R}^{d \times d},$$
$$\mathbb{T}_1^{(2)}(\mathbb{R}^d) \cong \exp\left(\mathbb{R}^d \oplus \mathbb{R}^{d \times d}\right).$$

Rough paths as Lie-group valued paths

- $\mathbb{T}_0^{(2)}(\mathbb{R}^d) \cong \mathbb{R}^d \oplus \mathbb{R}^{d\times d}, \quad \mathbb{T}_1^{(2)}(\mathbb{R}^d) \cong \exp\left(\mathbb{R}^d \oplus \mathbb{R}^{d\times d}\right)$:

  $\mathbb{T}_0^{(2)}(\mathbb{R}^d)$ is a Lie algebra with respect to the bracket

$$[b+c, b'+c'] = b\otimes b' - b'\otimes b.$$

- Let $\mathfrak{g}^{(2)} \subset \mathbb{T}_0^{(2)}(\mathbb{R}^d)$ be the sub-algebra generated by elements of the form $(0, b, 0)$:

  - $\mathfrak{g}^{(2)} = \mathbb{R}^d \oplus \mathfrak{so}(d) = \mathrm{span}\{e_i\} \oplus \mathrm{span}\{e_{ij} := [e_i, e_j], 1 \le i < j \le d\}$
  - $\mathfrak{g}^{(2)}$ is closed under $[\cdot,\cdot]$

  $G^{(2)}(\mathbb{R}^d) := \exp(\mathfrak{g}^{(2)})$ is a Lie sub-group of $\mathbb{T}_1^{(2)}(\mathbb{R}^d)$.

  :the step-2 nilpotent Lie group with d generators.

Remark. A rough path $(X, \mathbb{X}) \in \mathscr{C}^\alpha$ is geometric
if and only if the map $t \mapsto \mathscr{X}_t$ takes value in $G^{(2)}(\mathbb{R}^d)$.

Norm and Metric

- $G^{(2)}(\mathbb{R}^d)$ admits a natural *homogeneous Carnot-Carathéodory norm* $\|\cdot\|_C$ with the property, for $\mathbf{x} = b + c$,

$$\|\mathbf{x}\|_C \asymp |b| + |c|^{\frac{1}{2}},$$

where $\asymp$ Lipschitz equivalence with constants that may depend on the dimension d.

- It generates *homogeneous Carnot-Carathéodory metric* d_C such that

$$d_C(\mathbf{X_s}, \mathbf{X_t}) = \|\mathbf{X}\|_C \asymp |X_{s,t}| + |\mathbb{X}_{s,t}|^{\frac{1}{2}}.$$

Norm and Metric

- We define the *truncated signature* of a smooth path $\gamma : [0,1] \to \mathbb{R}^d$

by

$$G^{(2)}(\mathbb{R}^d) \ni S^{(2)}(\gamma) = \left(1, \int_0^t d\gamma(t), \int_0^1 \int_0^t d\gamma(s) \otimes d\gamma(t)\right),$$

then we have

$$\|\mathbf{x}\|_C := \inf \left\{ \int_0^t |\dot{\gamma}(t)| dt : \ \gamma \in \mathcal{C}^1([0,1]; \mathbb{R}^d), \ S^{(2)}(\gamma) = x \right\}.$$

Conclusion

Summary

1. For $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, the space $\mathscr{C}^\alpha([0,T], V)$ of α-Hölder rough paths is the space of pairs $(X, \mathbb{X}) \in \mathcal{C}^\alpha([0,T]; V) \oplus \mathcal{C}_2^{2\alpha}([0,T]^2; V \otimes V)$,

$$\|X\|_\alpha + \sqrt{\|\mathbb{X}\|_{2\alpha}} = \sup_{x \neq s \in [0,T]} \frac{|X_{s,t}|}{|t-s|^\alpha} + \left(\sup_{x \neq s \in [0,T]} \frac{|\mathbb{X}_{s,t}|}{|t-s|^{2\alpha}} \right)^{\frac{1}{2}} < \infty,$$

satisfying Chen's relation:

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = X_{s,u} \otimes X_{u,t}, \quad \forall s, u, t \in [0,T].$$

2. The space of geometric rough paths is

$$\mathscr{C}_g^\alpha([0,T], V) := \left\{ (X, \mathbb{X}) \in \mathscr{C}^\alpha : \mathrm{Sym}(\mathbb{X}_{s,t}) = \frac{1}{2} X_{s,t} \otimes X_{s,t} \right\}.$$

3. $(X, \mathbb{X}) \in \mathscr{C}_g^\alpha \iff$ the map $t \mapsto \mathscr{X}_t$ takes value in $G^{(2)}(\mathbb{R}^d)$.

Exercise

Exercise 2.9. (Interpolation)

Assume that $\mathbf{X^n} \in \mathscr{C}^\beta$, for $\beta \in (\frac{1}{3}, \frac{1}{2}]$, with uniform bounds

$$\sup_n \|X^n\|_\beta < \infty \quad \text{and} \quad \sup_n \|\mathbb{X}^n\|_{2\beta} < \infty$$

and pointwise convergence $X^n_{0,t} \to X_{0,t}$ and $\mathbb{X}^n_{0,t} \to \mathbb{X}_{0,t}$.

Show that this implies

$$\mathbf{X} \in \mathscr{C}^\beta, \quad \text{and} \quad \varrho_\alpha(\mathbf{X^n}, \mathbf{X}) \to \mathbf{0},$$

for every $\alpha \in (\frac{1}{3}, \beta)$.

THANK YOU FOR YOUR ATTENTION!